Lab 1

1. Write a program to print 'Hello, World!'

2. Write a program to take two integers as input and display their sum.

3. Write a program to find the largest of two numbers.

4. Write a program to check if a number is even or odd.

5. Write a program to calculate the factorial of a given number.

6. Write a program to print the first 10 natural numbers.

7. Write a program to find the sum of digits of a number.

8. Write a program to reverse a given number.

9. Write a program to check if a number is prime.

10. Write a program to display the multiplication table of a given number.

11. Write a program to swap two numbers without using a temporary variable.

12. Write a program to check if a string is a palindrome.

13. Write a program to find the largest element in an array.

14. Write a program to sort an array in ascending order.

15. Write a program to search for an element in an array (linear search).

16. Write a program to implement binary search on a sorted array.

17. Write a program to count vowels and consonants in a string.

18. Write a program to merge two arrays into a third array.

19. Write a program to find the Fibonacci series up to n terms.

20. Write a program to find both the minimum and maximum elements in an array.

LAB2

1. Write a program to print a message input from user (e.g., name of
your Institute).

2. Write a program to perform all basic arithmetic operations
(addition, subtraction, multiplication, division) on two numbers
and display answers of all operations.

3. Write a program to find the square and cube of a number.

4. Write a program to convert temperature between Celsius and

Fahrenheit.

5. Write a program to calculate simple and compound interest.

6. Write a program to swap two numbers using a temporary variable.

7. Write a program to print the ASCII value of a character.

8. Write a program to calculate the average of three numbers.

9. Write a program to find the area and perimeter of basic shapes

(rectangle, square, circle and triangle).

10.Write a program to find the volume of solid shapes (cuboid, sphere,

and cylinder).

LAB 3

1. Write a program to find whether a number is even or odd.

2. Write a program to check whether a given year is a leap year or not.

3. Write a program to find the greatest of three numbers.

4. Write a program to reverse a number.

5. Write a program to check whether a number is prime or not.

6. Write a program to check whether a number is an Armstrong number.

7. Write a program to check whether a number is a palindrome.

8. Write a program to calculate the sum of digits of a number.

9. Write a program to find the first and the last element in an array.

10. Write a program to declare and initialize variables of different data types (int, float, char,

double) and display them.

11. Write a program to print the size (in bytes) of all basic data types using sizeof().

12. Write a program to demonstrate implicit type conversion (e.g., int to float) in an arithmetic

expression.

13. Write a program to demonstrate explicit type conversion (type casting) in an arithmetic

expression.

14. Write a program to declare a const variable and show that its value cannot be changed.

15. Write a program to calculate and display the sum of elements in a float array (without using

loops, just by adding manually).

16. Write a program to declare and display a string using a character array.

17. Write a program to read a string from the user and display it back.

18. Write a program to demonstrate the use of escape sequences (\n, \t, \\, \").

19. Write a program to check whether a given character is uppercase or lowercase (without

using any predefined function).

20. Write a program to convert a lowercase character to uppercase (using ASCII value

manipulation).

21. Write a program to store and display the marks of 3 students using an array.

22. Write a program to declare and initialize a double variable and display it with 2 decimal places.

23. Write a program to declare and display values using the enum datatype.

24. Write a program to demonstrate the difference between int division and float division.

25. Write a program to declare variables using short, long, unsigned int and display their sizes and values.

LAB4

1. Find the factorial of a number entered by the user.

2. Find sum of odd and even digits of a number taken from user input.

3. Write a program that checks if a number is divisible by both 3 and 5.

4. Make a countdown timer using a while loop that counts down from 10 to 0.

5. Accept numbers from the user until zero is entered and display their sum.

6. Take user input marks (0–100) and print their grade: A (90+), B (80–89), C (70–79), etc.

7. Write a temperature checker that tells the user whether it's 'Hot', 'Cold', or 'Moderate'.

8. Create a basic login system with username and password and print Welcome message on

successful login.

9. Convert a binary number entered as an integer to its decimal equivalent.

10. Convert a decimal number entered as a float to its binary equivalent.

11. Accept a binary number as a string, and convert it to decimal.

12. Accept a decimal number and print its binary representation using bitwise operators.

13. Convert a decimal number to hexadecimal and vice versa.

14. Use Library functions to find and display the value of Pi, the square root, cosine, and

exponential functions for user input.

15. Use floor() and ceil() on a floating-point number and print both results.

16. Write a program that shows the diƯerence between local variable and static variable.

17. Write a program that demonstrates the use of an extern function declared in a header file

and used in the main file.

18. Format a string to display 5 student's details (name, roll number, percentage) in centred,

tabular format using escape sequences.

19. Create a formatted receipt showing item names, quantities, and prices aligned in

columns.

20. Accept a string and show diƯerent formatting styles, e.g., right-justified, left-justified, and

controlling decimal places in floats.

LAB 5

1. **Read a time input (hh:mm:ss) and print it with leading zeros if needed.**
2. **Display the memory address of an integer variable**
3. **Read an integer (where number of digits are greater than 4) and print first 4 of its digits in right-aligned width of 10**
4. **Accept a hexadecimal input (e.g., 0x1A) and print its decimal equivalent.**
5. **Read a sentence, count the number of words, digits, special characters and spaces.**
6. **Count the number of vowels and consonants in a string.**
7. **Check if a string is a palindrome.**
8. **Concatenate two strings**
9. **Find the frequency of each character in a string.**
10. **Remove all spaces from a string.**
11. **Accept a string and count how many times a substring occurs.**
12. **Accept a string and print the longest word.**
13. **Find the maximum and minimum elements in an array.**
14. **Find the average of array elements, where the array size is n.**
15. **Accept two arrays and merge them into one.**
16. **Accept a matrix and print it.**
17. **Remove duplicates from an array.**
18. **Find the sum of each row and column in a matrix.**
19. **Accept an array and find the sum of diagonal elements.**
20. **Accept an array and insert a new element at a given position.**

LAB 6

Patterns using Loops

1. Write a C program to print the following pattern:

*

* *

* * *

* * * *

* * * * *

2. Write a C program to print the following number pattern:

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

3. Write a C program to print the following pattern:

*

***

*****

****

****

4. Write a C program to print the reverse of the above triangle pattern:

****

****

*****

***

*

5. Write a C program to print a diamond pattern like this:

*

***

*****

***

*

1

6. Write a C program to print a diamond pattern like this:

****

*****

***

*

***

*****

****

7. Write a C program to print a diamond pattern like this:

* *

** **

*** ***

**** ****

*** ***

** **

* *

User Defined Functions

8. Write a function to sum array elements that are divisible by both 2 and 7.

9. Write an integer returning function with an integer argument that calculates how many left rota-

tions it takes for a number to return to its original state.

10. Create a function that monitors user input and counts vowels. If the vowel count exceeds 10, it

prints an alert. The count should be tracked across multiple calls.

11. Create an integer value holding buffer that will hold two values and lose old values on insertion of

new values. Print the lost values and total count of lost data on every insertion.

12. A cashier accepts coins (available in denominations of 1, 2, 5, 10) for a bill valued at 275. The

number of coins a customer will give is unknown. Write a function that keeps track of the total

coins and their cumulative sum, remaining change of coins and stops automatically.

User Defined Header Files

13. Create a header file that will contain two functions where one will reverse a string and another will

check if the reverse string is same as original.

14. A weather station logs hourly temperature. They want reports like max/min temperature of the

day. Write ”sensor.h” that will record temperature, find out maximum and minimum temperature.

15. A local library still uses pen-and-paper to track books. They want an automated system for the

staff to manage books. Create ”library.h” with functions that can add, remove and show all books

in library and use that header file.

16. A student wants to order food but he is unaware how much he will have to pay for a dish after

GST is applied, a discount coupon is used and delivery charges are added, which are calculated

based on how far he is from restaurant. Find out the final amount to be paid.